

Phase-1 App

The original end-to-end demo, deliberately frozen so it can never be quietly improved.

PORT	PAGE FILE	ENGINE	STATUS
8501	app/streamlit_app.py	MLP + LightGBM + climatology	FROZEN, read-only

Before anything clever was attempted, one question had to be answered: can a satellite surface picture be turned into a full temperature profile down to 1000 m at all? The Phase-1 App is the answer to that question, and it has been sealed shut ever since.

1. What this feature is, in plain words

The Phase-1 App is a small web dashboard. You open it in a browser, you pick a point in the North Indian Ocean and a date, and it draws the temperature of the water underneath that point at fifteen standard depths, from the surface down to one kilometre.

Underneath it are three separate methods that all answer the same question, so the answers can be compared: a small neural network that reads eleven surface numbers and writes out fifteen depth temperatures; a LightGBM model built as a deliberate rival; and plain climatology, which is simply the long-term average for that month. Climatology is the bar every model must clear - if a model cannot beat the average, it has learned nothing.

The important word in this note is **frozen**. This app is marked read-only in the project's own rules. No later work is allowed to edit it. That is not laziness; it is how we can still prove today what the system could genuinely do on the day it was first demonstrated, without anyone retrofitting a better result into it.

THE EVERYDAY PICTURE

Think of a doctor who can only take your skin temperature, but has to report your core temperature. Phase 1 is the first working version of that instrument for the ocean - rough, honest, and sealed in a glass case so nobody can adjust it after the fact.

2. What you actually see on the screen

Four panels, each one answering a different kind of question:

- **Map panel** - the whole basin at one chosen depth, coloured by temperature, so you see the shape of the water and not just a number.
- **Profile panel** - one point, fifteen depths, drawn as a vertical line down the page. This is the shape an oceanographer actually reads.
- **Priority panel** - the first version of 'where would another measurement be worth most', built from anomaly, uncertainty and float sparsity combined as a geometric mean.
- **Validation panel** - the model's own report card against real Argo floats it never saw during training.

3. How to run it, step by step

One command. It needs no GPU and no network; everything it reads is already on disk.

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/streamlit_app.py --server.port 8501
```

- 1 Open a terminal in the project folder and activate the project's virtual environment.
- 2 Run the command above. Streamlit prints a local address, normally **http://localhost:8501**.
- 3 Open that address in a browser. The map panel loads first.
- 4 Use the depth selector to move between the fifteen standard depths, and watch the basin change shape as you go deeper.
- 5 Switch to the profile panel and enter a latitude and longitude inside 5-30 N, 45-105 E to get the full column at that point.
- 6 Finish on the validation panel - that is where the model is shown being checked, which is the panel a jury will care about most.

IF A PANEL LOOKS PLAIN

The shell has a deliberate fallback: if a specialist panel is unavailable it draws a simpler built-in version rather than crashing. A demo that always renders something is worth more than one that renders beautifully or not at all.

4. Why the problem statement needs this

The problem statement asks for a system that reconstructs subsurface temperature from surface satellite observations and reports RMSE, correlation and bias. Phase 1 is the first complete pass through that entire chain - download, preprocess, train, predict, validate, display - with nothing faked in the middle.

It also settled the project's engineering rules, and those rules are why the later numbers can be trusted. Normalisation values ride inside the model file itself, so a checkpoint can never be scored with the wrong scaling. One data loader is shared by training and evaluation, so the two can never drift apart. The comparison between models uses a fixed tie rule agreed in advance, so nobody picks the winner after seeing the scores.

PS requirement	How this feature answers it
Reconstruct profile from surface	One forward pass returns all 15 standard depths for a chosen point and date.
Evaluate RMSE, correlation, bias	The validation panel reports all three, per depth, against independent Argo floats.
15 standard depths, 0-1000 m	The depth list is a frozen constant in <code>src/oceanembed/config.py</code> and is imported everywhere, never retyped.
Proof of concept over the NIO	The grid is exactly 5-30 N, 45-105 E at 0.25 degrees - $100 \times 240 = 24,000$ cells.

5. The measured numbers behind it

Measured against 879 independent Argo profiles that were never used in training [VERIFIED]:

What was measured	Value	Reading
RMSE against independent Argo	0.9638 degC	average error across all depths
Skill against climatology	+0.387	39% better than the long-term average
Training data	48 monthly GLORYS steps, 2019-2022	monthly, not daily - the Phase-2 limit this exposed
Model size	11 inputs -> 128 -> 128 -> 15 outputs	small on purpose

- It works, and it is modest. 0.9638 degC is a real result on real held-out floats, and it is the number Phase 2 was built to beat.
- The monthly cadence was the binding limit. Averaging a whole month erases cyclones, eddies and fronts - which is precisely the motivation for the daily Phase-2 system in Feature 7.

6. The impact - and who it is for

A first version that is honest about being a first version is worth more to a reviewer than a polished one with no history.

Who	What changes for them
The jury	A visible before-and-after. They can watch the project improve from 0.9638 to 0.9063 degC and see exactly what bought the improvement.
The sponsor (INCOIS / MoES)	Proof the full pipeline exists and runs on their region, their depths and their grid - not on a toy problem.
The team	A frozen reference point. Any later regression is instantly visible because this one cannot move.
A future maintainer	The engineering rules that keep the later numbers honest were all written here, and are documented as numbered decisions.

WHY SOCIETY GAINS FROM THIS

Every operational forecast system starts as somebody's first working version. Freezing it, publishing its worst numbers and refusing to edit it afterwards is the difference between a research demo and something a public agency could eventually rely on.

7. Say this to the jury

This is where we started, and we have not touched it since. It reconstructs the full water column from surface data, it scores 0.9638 degC against 879 floats it never saw, and it is locked read-only so you can see exactly how far the project moved afterwards. Everything else we will show you today had to beat this.

30-second opener - use this to set up the improvement story.

8. Questions the jury will ask

Q. Why show us an old version at all?

A. Because it is the control. Without it, our final number is just a number. With it, we can show a measured improvement from 0.9638 to 0.9063 degC on independent floats - different years, grids and scoring protocols, so a comparison of eras rather than a controlled one - and name what changed: daily data instead of monthly, and a 3-D encoder over an 11-day window instead of a per-column MLP. The climatology-anchored decoder we built was measured to cost accuracy and is NOT shipped.

Q. Is climatology not a very weak baseline?

A. It is the standard one in this field, and it is not weak everywhere. At 1000 m climatology still beats our shipped model by 0.024 degC. We label that on the chart rather than hide it - our model wins at 14 of 15 depths, not 15.

Q. How do you know the floats were really held out?

A. The Argo profiles are never in the training set, and the train/test split is a frozen constant in the config file with an assertion that runs at import time in the tests. The split cannot be changed by editing a script.

9. The honest limit - say it before they find it

TWO KNOWN PROBLEMS, BOTH STATED

The Phase-1 out-of-distribution detector is not validated - it flags 99.18% of the real test set, because the reference file it compares against is a stale synthetic one. The detector logic is correct; its input file is wrong. Separately, the Phase-1 MC-dropout uncertainty was measured overconfident at every depth, worst at 3.54x at 20 m. Both are recorded in the project decisions log, and neither is used by the shipped Phase-2 system, which has its own calibrated uncertainty - see Feature 13.

IN ONE LINE

The first complete surface-to-subsurface system, scored on independent floats and then sealed read-only so the improvement that followed is measurable rather than asserted.

0.9638 degC RMSE VS 879 INDEPENDENT FLOATS	+0.387 SKILL OVER CLIMATOLOGY	FROZEN READ-ONLY BY PROJECT RULE
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Feature 1 of 17. Next: Feature 2.